

A Comparative Analysis of Hydraulic Stability Methods for Rock Protection of Subsea Cables in Nearshore Environments

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Content

1. Introduction	3
2. The Nearshore Environment	4
2.1 Definition and area of interest.....	4
2.2 Complexity of Nearshore Environment.....	5
2.3 Bed Morphology and Composition	8
3. Hydrodynamic Forcing	10
3.1 Currents	10
3.2 Waves.....	12
4. Stability Parameters & Criteria	13
4.1 The Shields Parameter (Uniform Flow)	14
4.2 Turbulence-based Parameters.....	14
4.3 Combined Wave-Current Conditions	15
4.4 Decision Framework	15
5. Rock Stability Assessment Methods.....	16
5.1 Hudson Formula	16
5.2 Van der Meer (1988/1998)	17
5.3 Van Rijn (2019)	18
5.4 Van Gent & Van der Werf (2014)	19
5.5 Herrera & Medina (2015)	20
5.6 Deltares HaSPro Handbook (2023)	21
5.7 Rock Density	23
5.8 Comparative Summary	24
6. Scour	25
6.1 The Scour Process.....	25
6.2 Scour Around Subsea Cables	25
6.3 Scour and Berm stability.....	26
6.4 Scour Mitigation Approaches	26
6.5 Anchor and Trawling Protection.....	27
7. Design Framework & Computational Approach.....	29
7.1 Input Parameter Space	29
7.2 Core Design Formulae.....	29
7.3 Failure Criteria & Damage Levels	29
7.4 Validity Ranges & Applicability Warnings.....	29
7.5 Sensitivity Analysis Framework	29

8.	Knowledge Gaps & Research Questions	31
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1. Introduction

- The rapid growth of the offshore wind sector
- the increasing deployment of subsea cables as critical energy infrastructure
- the intensification of hydrodynamic loading due to climate change.
- Why are existing design methods are not always suitable for nearshore cable protection?
- the gap this thesis addresses.
- research objective and scope of the study.

Schiereck & Verhagen (2019) — Ch. 1 (Introduction: why and when, protection against what, failure and design)

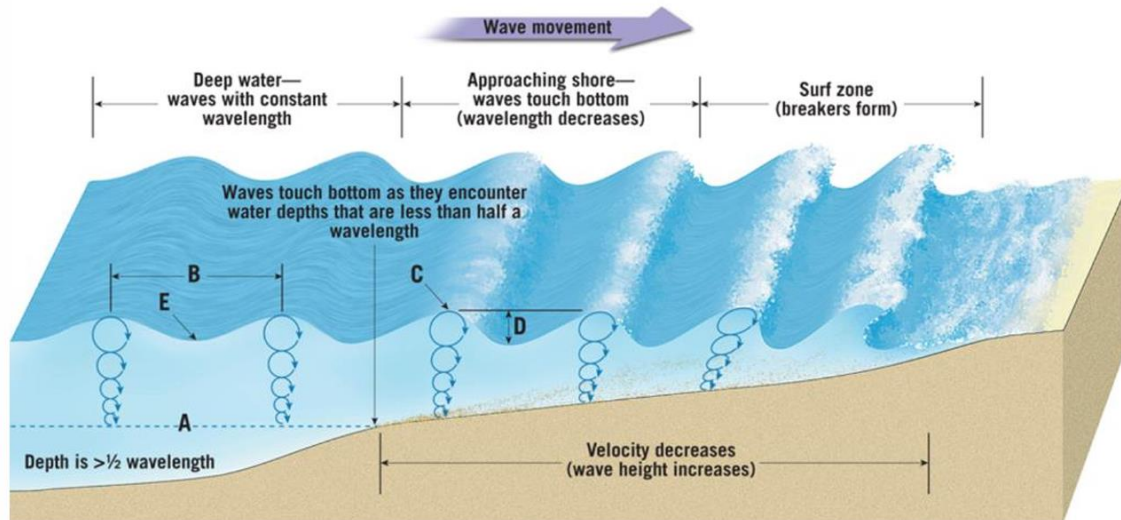
Deltares Handbook (2023) — Ch. 1 (Background, considered foundation types) & Ch. 3 (Scour mitigation strategies)

Nowadays we over dimension a lot, especially when a company designs the rock berm when they are responsible. They consider for everything the worst-case scenario, what results in a not sustainable design.

2. The Nearshore Environment

2.1 Definition and area of interest

The coastal region is not a single uniform zone but different areas with different physical difficulties, each characterized by different hydrodynamic conditions and levels of seabed interaction. For the purposes of this study, it is essential to distinguish between three zones: the deepwater/offshore zone, the nearshore zone/approaching zone, and the surf zone (Figure x).



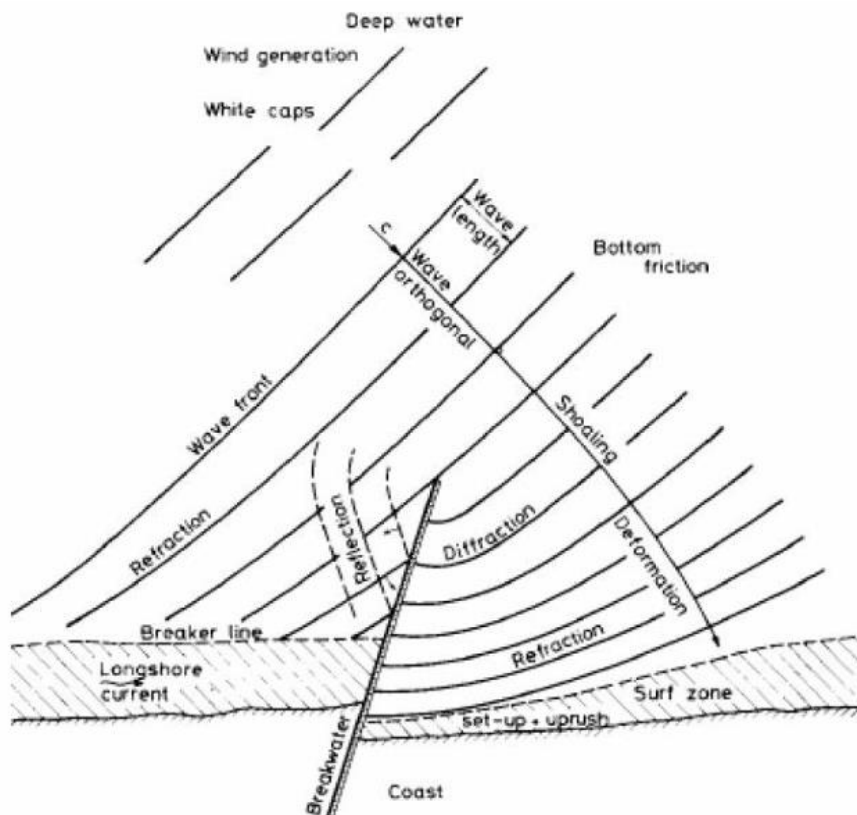
The offshore zone is characterized by water depths sufficiently large that surface waves do not interact with the seabed. In this region, waves propagate largely undisturbed, and the flow conditions above the bed are governed primarily by tidal and large-scale oceanic flows. Wave-induced orbital velocities become negligible at the seabed when the water depth exceeds approximately half the wavelength ($h > L/2$). Offshore conditions are therefore relatively well understood and predictable, and the design of cable protections in this zone is more straightforward.

The nearshore zone begins where waves start to feel the seabed, where ($h < L/2$) and extends landward to the point where waves begin to break consistently. This is sometimes referred to as the shoaling zone. In this zone, the interaction between wave orbital motion and the seabed becomes significant: wave celerity decreases, wavelength shortens, and wave height first increases through shoaling before eventually being limited by depth-induced breaking. This zone is the primary focus of this study, as it is precisely here that many nearshore cable routes are laid, and where the complexity of hydrodynamic loading is greatest.

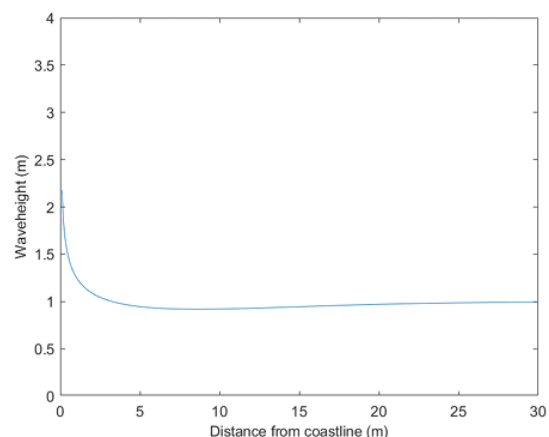
The surf zone extends from the breaker line to the shoreline. In this zone, waves have broken and energy dissipation through turbulence dominates. Longshore currents, undertow, and swash processes are characteristic of this zone. While surf zone dynamics are of interest to coastal engineers, subsea cable routes generally avoid the surf zone due to the extreme and highly variable loading conditions. This study therefore focuses on the nearshore zone, with some attention to the transition region near the breaker line where also wave breaking can occur.

2.2 Complexity of Nearshore Environment

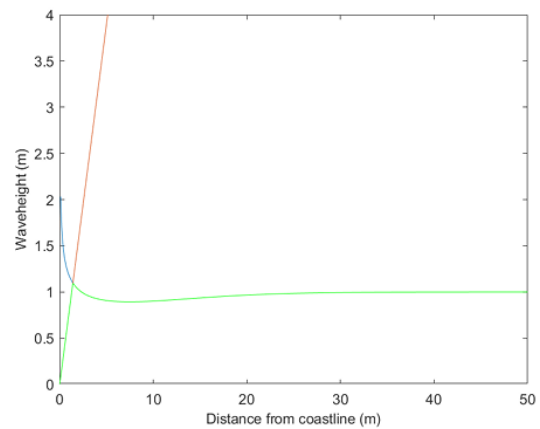
What makes the nearshore zone fundamentally more complex than the offshore environment is the convergence of multiple physical processes acting simultaneously at or near the seabed. The following processes are of direct relevance to the hydraulic stability of rock protection systems.



Shallow water effects. As water depth decreases, the behaviour of waves departs progressively from deep-water linear wave theory. The dispersion relation becomes depth-dependent, wave orbital velocities at the seabed increase significantly, and the wave profile becomes asymmetric, with steeper fronts and flatter troughs. This asymmetry leads to net sediment transport in the direction of wave propagation, which is an important consideration for the long-term stability of any bed protection. Schiereck and Verhagen (2019) describe how waves travelling from a depth of 20 m towards the shore undergo a complex sequence of shoaling, bottom friction, and eventually breaking, with the relative importance of each process depending strongly on the bed slope.



Wave breaking. Breaking occurs when a wave can no longer sustain its form, either because it becomes too steep in deep water or because the water depth becomes too shallow. The breaking criterion given by Miche provides a theoretical upper limit for wave height as a function of local depth and wavelength (Schierreck & Verhagen, 2019, Eq. 7.8). In practice, for shallow foreshores, a useful approximation is $H_b/h_b \approx 0.8-0.88$, indicating that the significant wave height is strongly depth-limited in the nearshore zone.

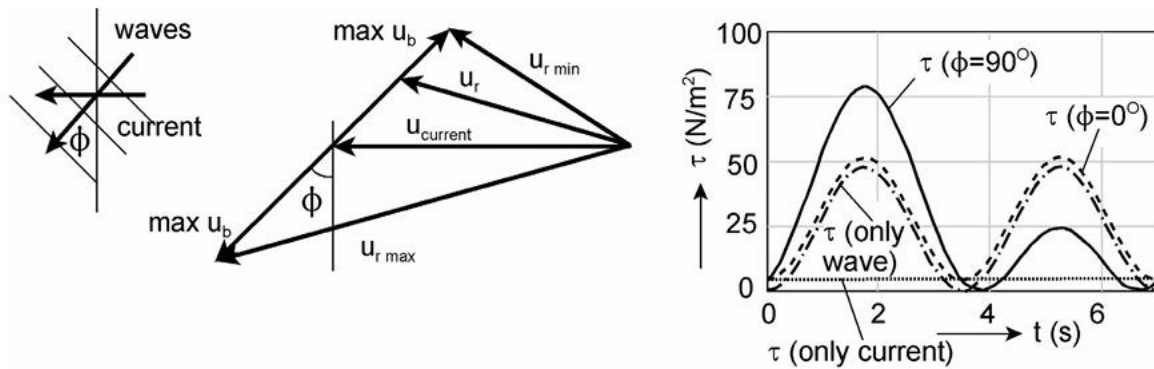


Breaking waves represent a fundamentally different loading category compared to non-breaking waves. Before breaking, the wave field is oscillatory and can be reasonably described by linear wave theory and standard transformation methods. After breaking, the flow becomes turbulent and impulsive, and the impact forces generated are difficult to predict using standard wave theories. For rock protection design, this means that stability formulae derived for non-breaking or mildly breaking conditions may significantly underestimate the actual loading (Herrera & Medina, 2015).

This distinction directly affects which stability method is applicable in this thesis (Chapter 6). Herrera and Medina (2015) specifically address toe berms in very shallow water on steep slopes under breaking wave conditions, making their method one of the few in this comparison validated for the breaking regime. Van Gent and Van der Werf (2014) and Van Rijn (2019), by contrast, are primarily validated for non-breaking or mildly breaking conditions; their applicability must therefore be checked explicitly wherever a design point along the cable route falls within, or close to, the breaking zone. Because the position of the breaker line shifts with the tidal cycle (see below), a single design point may alternate between breaking and non-breaking loading regimes over time.

Tidal influence. Tidal water-level variation causes the breaker line, and therefore the boundary between breaking and non-breaking loading, to shift back and forth across the seabed over the tidal cycle. As a result, a protection layer that is stable under non-breaking wave action at high water may be exposed to breaking wave impact at the same location during low water. A better treatment of tidal currents themselves is given in Chapter 3 (Currents), where they are treated as context rather than a governing load.

Wave-current interaction. Waves and currents generally co-exist in the nearshore zone, and their interaction is non-linear: the combined bed-shear stress produced by simultaneous waves and current exceeds the simple sum of each contribution separately, and depends on the angle between wave and current direction (Schierreck & Verhagen, 2019, Section 7.2.2). Consistent with the scope of this thesis, waves are treated as the governing load; a full treatment of combined wave-current loading is kept open as a possible extension (see Chapter 3).



Sediment dynamics. Unlike the offshore zone where the seabed may be relatively stable, nearshore beds are often composed of mobile sediments, fine to medium sands, that are in a state of quasi-continuous transport under the combined action of waves and currents. This has direct consequences for rock protection: sediment can be winnowed (interface instability or piping) out from beneath or between the protection stones. The protection layer can sink into a softer underlying layer, and erosion around the edges of a protection berm can weaken its stability. The Deltares HaSPro Handbook (2023) states that, besides external stability, interface stability and flexibility are two of the three main requirements for offshore rock protections.

- Schiereck & Verhagen (2019)

Ch. 7.3 (Breaking waves, waves on a foreshore)

Ch. 2 (Flow loads, introduction)

- Deltares Handbook (2023)

Ch. 2 (General definitions, general description of the scour process)

2.3 Bed Morphology and Composition

Subsea cable routes in nearshore environments have a lot of variety of seabed types, each presenting distinct challenges for rock protection design. Two of these conditions are treated as the governing design cases in this thesis: bedrock, where trenching is not possible due to the hardness of the formation, and mobile sandy seabed, encountered mainly at crossings and around obstacles where burial is otherwise not feasible.

Rocky and consolidated beds are commonly found along steep coastlines, especially in fjord areas and around rocky headlands. These seabeds are generally stable and do not carry a risk of winnowing. However, their uneven surface can make it difficult for the protection berm to fully connect with the seabed, which may create unsupported gaps beneath the berm. Herrera and Medina (2015) explain that steep rocky coastlines (with slopes of 1/10 or steeper) behave very differently from gentle sandy slopes, as a result, design formulae developed for sandy coastlines are not directly suitable for these conditions.

Sandy beds are the most common substrate in nearshore zones along low-lying coasts, and become the governing design case at crossings and around obstacles where burial is not feasible. Fine to medium sands (D50 in the range of 0.1–0.5 mm) are highly mobile under wave and current action, making them susceptible to scour around and beneath cable protection structures. Sandy beds also present challenges for filter design, as fine sediment can be transported through the voids of the protection layer if the underlayer grading is not properly designed (Schierbeck & Verhagen, 2019, Ch. 6), directly relevant to the interface stability requirement of this thesis.

Mixed sand-gravel and gravel beds occur in higher-energy, somewhat steeper nearshore environments, such as exposed Atlantic coasts or tidal inlets, these are generally more stable than pure sand beds but complicate the bed-protection interface due to their coarser, irregular texture. Morphological features such as sand bars, tidal channels, and cross-shore profiles with alternating shallow and deeper areas also occur along cable routes and create local variation in wave breaking, current strength, and bed-shear stress. These conditions are not treated as separate design cases in this thesis but are noted as local variations of the two governing seabed types above, which may require a more detailed spatial assessment along specific sections of the route.

The nearshore environment is physically complex because waves, tidal currents, wave breaking, wave-current interaction, and moving sediments all occur at the same time. These processes interact with each other and can create loads that are much larger than those caused by a single process alone.

The seabed in the nearshore zone can vary from mobile sandy beds to steep rocky areas, and each type creates different challenges for the design and long-term stability of rock protection systems. A good understanding of these conditions is necessary to determine whether existing stability methods, mainly developed for offshore or sheltered coastal environments, can also be applied to nearshore cable protection.

The following chapters build on this background step by step: Chapter 3 discusses hydrodynamic forcing, Chapter 4 introduces the stability parameters used in this thesis, including near-bed flow conditions, and Chapter 5 evaluates the available assessment methods.

- Schiereck & Verhagen (2019)

Section 7.3 (Breaking waves)

Section 7.3.2 (Waves on a foreshore)

Section 7.2.2 (Shear stress under waves)

Section 2.2 (Turbulence)

Ch. 6 (Filters)

- Herrera & Medina (2015)

Introduction and Section 1 (steep sea bottoms, shallow water)

- Deltares HaSPro Handbook (2023)

Ch. 2 (General definitions)

Ch. 4.2 (Performance requirements: external, interface, flexibility)

3. Hydrodynamic Forcing

3.1 Currents

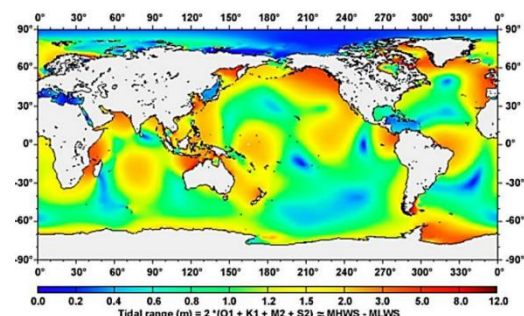
Currents in the nearshore zone are generated by a limited number of physical processes that influence the stability of rock protection along subsea cable routes. In this thesis, the nearshore is defined as the area seaward of the wave-breaking zone, where tidal flow and non-breaking wave motion are the main hydrodynamic processes of interest. These currents are important because they create bed shear stress, transport sediment, and can contribute to scour at the edge of the protection layer. For this reason, understanding the current types present and their typical magnitudes is essential for any stability assessment.

In hydraulic design, flow above a protection layer is often described using the depth-averaged velocity. This is a practical parameter because it is easy to use in calculations and design formulae. However, as Schiereck and Verhagen emphasise, the depth-averaged value alone is not enough for protection design. Stone stability also depends on the near-bed velocity, the shape of the vertical velocity profile, and the turbulence intensity close to the seabed. In nearshore conditions, these factors can differ significantly from the depth-averaged flow, especially where tidal flow and wave motion interact.

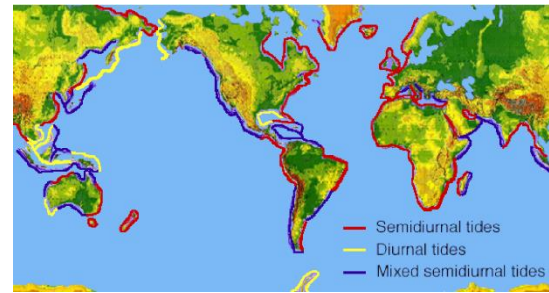
Tidal currents are the most important current mechanism in many nearshore environments and are highly relevant to subsea cable protection design. They are generated by the changing water level caused by the gravitational attraction of the Moon and the Sun acting on the rotating Earth. This astronomical forcing produces a periodic rise and fall of water level, and the horizontal movement of water required to follow that rise and fall creates the tidal current.

The main feature of tidal currents is that they reverse direction during each tidal cycle. During the flood phase, water moves landward; during the ebb phase, it moves seaward. This reversing pattern is important for rock protection because it can cause loading from both directions and may lead to alternating scour on opposite sides of the protection layer. If the flood and ebb phases are not equal, tidal asymmetry may occur, which can produce a net transport of sediment over time and gradually disturb the area around the protection berm.

The magnitude of tidal currents depends on the local tidal range, water depth, and coastal geometry. In some nearshore areas the tidal current is modest, while in other locations it can become strong enough to be one of the governing loads on the protection layer.

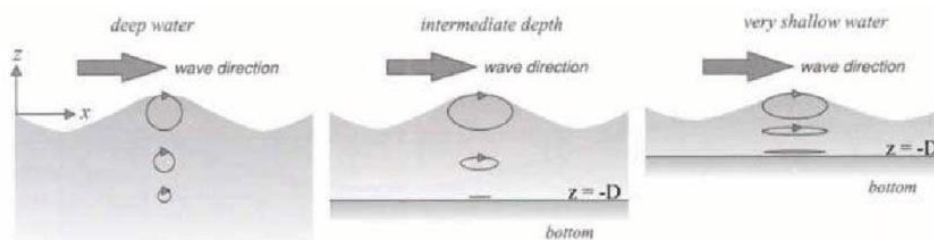


The tidal pattern at a location can also be semidiurnal, diurnal, or mixed, which affects how often the current reverses and how the load is distributed over time: a semidiurnal tide reverses twice a day in a regular pattern, a diurnal tide reverses once a day with longer current phases, and a mixed tide, with unequal flood and ebb, can produce an asymmetric scour pattern around the protection layer.



Wind-driven currents can modify the local current field, particularly when they reinforce the tidal current, though in the nearshore they are generally a secondary contribution that becomes more relevant during storm conditions. Density-driven currents, caused by differences in salinity or temperature, are generally a minor and site-dependent contribution to the total current field.

Wave-induced currents are more directly relevant, since wave motion produces oscillatory flow at the seabed even without breaking, and this motion can interact with the tidal current. These wave-related currents do not act like steady currents but produce repeated oscillatory motion that can destabilise stones, and in nearshore protection design the combined effect of waves and tidal current is often more important than either process alone.



Where waves break at an angle to the shoreline, the resulting alongshore momentum transfer generates a longshore current in the surf zone, whose direction follows the angle of wave incidence at breaking; this current is also the main driver of longshore sediment transport (Battjes, 1974, in Schiereck & Verhagen, 2019). Since cable routes generally avoid the surf zone itself, this mechanism is most relevant in the transition region near the breaker line discussed in Chapter 2, where it can influence sediment movement around the protection berm. **DNV?!**

For protection design, currents should ultimately be characterised in terms of their effect on the rock layer: near-bed velocity, direction, reversal, and the combined action with waves matter more than the depth-averaged velocity alone. Consistent with the scope of this thesis, the wave parameters used in the stability assessment (Section 3.2) are taken directly from NKT route data, and waves are treated as the governing load; a full characterisation of currents, and of combined wave-current loading beyond the qualitative description above, is therefore kept open as a possible extension of this work.

3.2 Waves

Waves are the governing hydrodynamic load considered in this thesis, since they generate oscillatory motion throughout the water column and, in particular, near the seabed. For each location along the cable route, NKT route data directly provide the significant wave height, the peak period, and the wave direction, so these values are used as input to the stability assessment without a separate derivation step.

The significant wave height, H_s , represents the mean height of the highest one third of the waves in a sea state and is used as the main measure of wave severity. The peak period, T_p , corresponds to the period at the peak of the wave spectrum, representing the frequency band containing the most wave energy, and together with H_s determines the near-bed orbital motion that loads the protection layer. Wave direction is retained as a design parameter because it determines whether wave attack is aligned with or oblique to the cable route, which affects both the local loading and the potential interaction with any current.

Because the wave climate at the location of interest is already available from the route data, the transformation steps that would otherwise be required, such as shoaling, refraction and numerical propagation with a model such as SWAN, are not needed in this thesis. Wave breaking, which does require separate attention because it changes the loading regime, is treated in Chapter 2.

This chapter has set out the hydrodynamic context for the stability assessment. Currents contribute to the loading on a protection layer, but the wave climate, taken directly from NKT route data, is treated as the governing load in this thesis. Chapter 4 introduces the stability parameters used to translate this loading into a stability criterion, including near-bed flow conditions, and Chapter 5 evaluates the available assessment methods.

- Schiereck & Verhagen (2019), Ch. 7 (Waves), Ch. 2 (Flow loads)
- Van Rijn (2019), Section 2 (Critical bed-shear stress, combined current and wave conditions, Eq. 5)
- Deltares Handbook (2023), Ch. 2 (Scour prediction, general description of the scour process)

4. Stability Parameters & Criteria

Rock stability is ultimately governed by the flow conditions experienced directly at the bed, not by the depth-averaged flow. In uniform, stationary flow, the boundary layer is fully developed and occupies the entire water depth, producing a logarithmic velocity profile with the average velocity located at approximately 0.4 times the water depth above the bed (Schierack & Verhagen, 2019, Ch. 2). Tidal currents, because of their long period relative to the time needed for the boundary layer to develop, can reasonably be treated as uniform flow for design purposes.

In practice, however, flow in the nearshore is rarely uniform. Accelerations and decelerations, caused for example by structures, bed irregularities, or wave orbital motion, change the thickness of the boundary layer and the shape of the velocity profile: accelerating flow produces a fuller profile with increased near-bed shear stress, while decelerating flow thickens the boundary layer and increases turbulence (Schierack & Verhagen, 2019, Ch. 2.3, 3.4). This distinction between uniform and non-uniform flow matters because the standard Shields approach (Section 4.1) is strictly valid only for the uniform case.

Turbulence itself is characterised by the turbulent kinetic energy, k , and by the relative turbulence intensity, defined as the ratio of the root-mean-square velocity fluctuation to the local mean velocity (Schierack & Verhagen, 2019, Ch. 2.2). Elevated turbulence, whether generated by waves, bed roughness, or nearby structures, increases the instantaneous forces on individual stones beyond what the mean velocity alone would suggest, which is why standard, mean-velocity-based stability criteria become less reliable in non-uniform or high-turbulence conditions. This is the basis for the turbulence-based stability parameter introduced in Section 4.2.

4.1 The Shields Parameter (Uniform Flow)

The classical stability criterion for granular material in a current is the Shields (1936) curve, which expresses the critical dimensionless bed-shear stress, or critical Shields mobility number, as a function of a dimensionless particle Reynolds number. For coarse material ($D_{50} > 10$ mm), such as the rock sizes considered in this thesis, the critical Shields parameter is approximately constant at $\theta_{cr,shields} = 0.05$, independent of the Reynolds number (Van Rijn, 2019).

The Shields data show considerable scatter, partly because the precise definition of "initiation of motion" is not sharply defined and partly because of experimental variability. For design purposes, Van Rijn (2019) represents this scatter with a reduction coefficient, r , applied to the standard Shields value:

$$\theta_{cr} = r \cdot \theta_{cr,shields} \quad \text{with } \theta_{cr,shields} = 0.05 \text{ and } r \text{ in the range of } 0.4 \text{ to } 1$$

The reduction coefficient r functions as a damage or safety parameter: a lower value of r corresponds to a more conservative design, while a value closer to 1 corresponds to the classical Shields threshold. This is the mechanism through which an agreed damage level can be translated into a required stone size, and it is used throughout the comparison in Chapter 5.

The Shields approach is reliable for uniform, current-only flow on a horizontal or mildly sloping bed. It becomes less reliable where the flow is non-uniform or where turbulence levels are elevated (Section 4.2), or where waves are present in addition to a current (Section 4.3).

4.2 Turbulence-based Parameters

Hofland (2005, as cited in Van Rijn, 2019) studied the stability of rock in non-uniform flows with relatively high turbulence levels and found that the local mean velocity alone is not sufficient to characterise the loading on individual stones in these conditions. Hofland introduced an additional stability parameter based on the turbulent kinetic energy, k , alongside the local mean velocity, u , to account for the extra destabilising effect of velocity fluctuations that the standard Shields approach does not capture.

This turbulence-based approach becomes necessary wherever the flow departs significantly from the uniform case in Section 4.1: near structures, at bed irregularities, or in the wave boundary layer. Because a free-standing rock berm itself introduces a bed irregularity, and because wave orbital motion in the nearshore is inherently non-uniform, turbulence effects are a relevant consideration for the seabed types and loading conditions in this thesis, alongside the combined wave-current approach of Section 4.3.

4.3 Combined Wave-Current Conditions

Where a current and waves act simultaneously, Van Rijn (2019) shows that the Shields approach remains valid provided the bed-shear stress is computed as the sum of a current-related and a wave-related contribution:

$$\tau_{b,cw} = \tau_{b,c} + \tau_{b,w} \text{ (Van Rijn, 2019, Eq. 5)}$$

with:

$\tau_{b,c} = 0.125 \cdot \rho_w \cdot f_c \cdot (\gamma_{str} \cdot u_c)^2$, the bed-shear stress due to current

$\tau_{b,w} = 0.25 \cdot \rho_w \cdot f_w \cdot (\gamma_{str} \cdot U_w)^2$, the bed-shear stress due to waves

where u_c is the depth-averaged current velocity, U_w is the near-bed peak orbital velocity from linear wave theory, and γ_{str} is a velocity and turbulence enhancement factor accounting for the presence of a structure such as a rock berm. The friction factors f_c and f_w depend on the relative roughness of the bed; Van Rijn (2019) also gives simplified power-function approximations for f_c and f_w to avoid the iterative solution that the standard formulations require.

This method is directly applicable to the scope of this thesis, since it is, among the five core sources, the only one that explicitly covers combined wave-and-current loading on a bed or berm.

4.4 Decision Framework

The choice between the parameters introduced in Sections 4.1 to 4.3 depends on the hydrodynamic conditions at a given point along the cable route:

- Uniform current, no waves: the Shields parameter (Section 4.1) applies directly.
- Non-uniform flow or elevated turbulence, without a dominant wave component: the turbulence-based approach (Section 4.2) is more appropriate.
- Wave-dominated conditions: methods developed specifically for wave loading, such as Van Gent and Van der Werf (2014) or Herrera and Medina (2015), are used (Chapter 5).
- Combined current and waves: the Van Rijn (2019) approach (Section 4.3) is used, since it is the only method in this comparison that explicitly covers this case.

This decision framework is central to the design tool described in Chapter 7: for a given combination of seabed type, wave and current conditions, and location along the route, the tool applies the parameter, and therefore the stability method, that is actually valid for those conditions.

- Schiereck & Verhagen (2019), Ch. 2.2 (Turbulence), Ch. 2.3 (Wall flow), Ch. 3.2 (Uniform flow), Ch. 3.4 (Non-uniform flow), Ch. 8.3 (Stability of loose grains under wave loading)
- Van Rijn (2019), Section 1 (Introduction), Section 2 (Critical Shields mobility parameter, Eq. 1–5, reduction coefficient r , Table 1)
- Deltares Handbook (2023), Ch. 4.2.1 (External stability of loose rock scour protections), Ch. 4.4.1 (External stability, deformation model, mobility limits)

5. Rock Stability Assessment Methods

The core of this chapter is a comparative analysis. For each method, this thesis considers: the physical assumptions it is based on, the required input parameters, the output quantity it produces, the damage level it is calibrated against, and its validity limits. These points are compared directly in Section 5.8.

5.1 Hudson Formula

The Hudson formula (Hudson, 1958) is the classical method for the stability of rock armour on a slope under wave attack, and remains widely used as a first estimate. It relates the required stone weight to the wave height, the relative buoyant density, the slope angle, and a stability coefficient KD. KD is essentially a "dustbin factor" that implicitly absorbs the accepted degree of damage, material type, and placement method, and takes different values for different armour units, roughly 3–4 for natural rock (Schierack & Verhagen, 2019, Ch. 8).

The formula's simplicity comes at a cost: it does not include wave period or a current, and KD is calibrated for approximately 5% damage without giving an explicit damage scale, so results cannot be scaled to other damage levels. Hudson's formula is validated primarily for slope angles in the range $1.5 < \cot\alpha < 4$, and for waves that do not break at or overtop the toe of the slope. Because it does not represent a bed or berm on a horizontal or mildly sloping seabed, its role in this thesis is limited to a quick, conservative first estimate rather than a design method (Section 5.8).

$$Dn50 = H_s / (\Delta \cdot (KD \cdot \cot\alpha)^{1/3})$$

Variable	Meaning	How to determine
Dn50	Required nominal stone diameter (m)	Output
Hs	Significant wave height (m)	Route data
Δ	Relative buoyant density, $(\rho_s - \rho_w)/\rho_w$	$\rho_s = 2650$ or 3000 kg/m^3 (variable)
KD	Stability coefficient	Table value, $\approx 3\text{--}4$ for natural rock
$\cot\alpha$	Slope gradient	Project/berm geometry

5.2 Van der Meer (1988/1998)

Van der Meer (1988) extended the Hudson approach for rubble-mound slope armour under random waves, introducing the wave period, the number of waves, and the notional permeability of the structure, P , as explicit parameters, with separate expressions for plunging and surging breakers. Unlike Hudson's implicit damage level, Van der Meer expresses damage explicitly through the damage number $S_d = Ae/Dn50^2$, the eroded cross-sectional area normalised by the square of the nominal stone diameter; a damage level of $S_d \approx 2-3$ is generally taken as the start of damage, and failure (where the armour layer is locally removed and the filter becomes exposed) at higher S_d values that depend on the slope (Schierreck & Verhagen, 2019, Ch. 8).

Because the equations are based on many physical model tests, they represent a fitted average of scattered data, with relative standard deviations of about 6.5% (plunging) and 8% (surging) that can be used directly in a probabilistic design. Like Hudson, however, Van der Meer's formulae were developed and validated for breakwater armour slopes, not for a berm on a horizontal or mildly sloping bed, and they do not include a current. Their relevance to this thesis is therefore mainly as a wave-only, slope-based reference point rather than a method for the free-standing berm case (Section 5.8).

$$\begin{aligned} \text{Plunging breakers: } Hs/(\Delta \cdot Dn50) &= 6.2 \cdot P^{0.18} \cdot (S_d/\sqrt{N})^{0.2} \cdot \xi m^{-0.5} \\ \text{Surging breakers: } Hs/(\Delta \cdot Dn50) &= 1.0 \cdot P^{-0.13} \cdot (S_d/\sqrt{N})^{0.2} \cdot \sqrt{\cot \alpha} \cdot \xi m^P \end{aligned}$$

Variable	Meaning	How to determine
S_d	Damage number = $Ae/Dn50^2$	Chosen damage level (start $\approx 2-3$, failure $\approx 8-17$ depending on slope)
N	Number of waves in the storm	From storm duration / T_m
P	Notional permeability	Table value depending on core structure (0 = impermeable, 0.6 = very permeable)
ξm	Iribarren number = $\tan \alpha / \sqrt{(Hs/Lm)}$	From slope and wave steepness
L_m	Wavelength at T_m	Linear wave theory

5.3 Van Rijn (2019)

Van Rijn (2019) extends the classical Shields approach (Chapter 4) specifically to the design of bed and berm protection for large stones, cobbles, boulders, and rocks in the range $D50 = 0.03$ to 3 m, under a current, waves, or both together. The method uses the critical Shields mobility number, $\theta_{cr} = r \cdot \theta_{cr, shields}$, shields with $\theta_{cr, shields} = 0.05$, where the reduction coefficient r (0.4 to 1.0) sets the accepted level of movement or damage, and combines current- and wave-related bed-shear stress as $\tau_{b, cw} = \tau_{b, c} + \tau_{b, w}$ (Chapter 4, Section 4.3).

Among the five core sources for this thesis, Van Rijn (2019) is the only method that explicitly covers combined wave-and-current loading on a horizontal or mildly sloping bed or berm, rather than a sloped breakwater armour layer, which makes it directly applicable to the free-standing rock berm considered here. It is calibrated against a compilation of field and laboratory data, though this field calibration is comparatively sparse for the largest rock sizes. The method supports both a static design (no movement allowed) and a dynamic design (some permitted movement via the reduction coefficient r), which is useful for the "avoid overdesign" objective of this thesis.

$$\begin{aligned}\theta_{cr} &= r \cdot \theta_{cr, shields}, \theta_{cr, shields} = 0.05, r = 0.4 - 1.0 \\ \theta &= \tau_{b, cw} / (\Delta \cdot \rho_w \cdot g \cdot D50) \leq \theta_{cr} \rightarrow D50 = \tau_{b, cw} / (\Delta \cdot \rho_w \cdot g \cdot \theta_{cr}) \\ \tau_{b, cw} &= \tau_{b, c} + \tau_{b, w} \\ \tau_{b, c} &= 0.125 \cdot \rho_w \cdot f_c \cdot (\gamma_{str} \cdot u_c)^2 \\ \tau_{b, w} &= 0.25 \cdot \rho_w \cdot f_w \cdot (\gamma_{str} \cdot U_w)^2 \\ U_w &= \pi \cdot H_s / (T_p \cdot \sinh(2\pi h / L_s)) \\ f_c &= 0.24 / [\log(12h / k_s)]^2 \text{ (simplified: } f_c, a = 0.11 \cdot (h / k_s)^{-0.3} \text{)} \\ f_w &= \exp\{-6 + 5.2 \cdot (A_w / k_s)^{-0.19}\} \text{ (simplified: } f_w, a = 0.1 \cdot (A_w / k_s)^{-0.3} \text{)} \\ A_w &= (0.5 \cdot T_p / \pi) \cdot U_w\end{aligned}$$

Variable	Meaning	How to determine
uc	Depth-averaged current velocity (m/s)	Current data (Chapter 3), if applicable
h	Water depth (m)	Bathymetry / route data
Ls	Wavelength at significant wave (m)	Linear wave theory, from T_p and h
ks	Nikuradse bed roughness (m)	$k_s = \alpha \cdot D50$, $\alpha = 1.5-2$ for narrowly graded rock
γ_{str}	Velocity/turbulence enhancement factor due to structure	1 without structure, higher with a berm present
ρ_w	Water density (kg/m ³)	≈ 1025 for seawater

5.4 Van Gent & Van der Werf (2014)

Van Gent and Van der Werf (2014) developed a stability formula specifically for the toe structure of a rubble-mound breakwater, based on the characteristic near-bed orbital velocity, $\hat{u}$, rather than the wave height directly. The formula explicitly includes the toe berm geometry, the berm width B_t and thickness t_t , and uses the damage number Nod (the number of displaced stones within a strip as wide as D_{n50}) as its damage criterion. It was validated for slopes of 1:1.5 and 1:2, for toe structures within 10% to 30% of the water depth in front of the structure (h_t/h), and for a water depth in front of the toe of 1.2 to 4.5 times the wave height (h/H_s); a roughness correction factor is also available for smoother armour slopes.

Because the formula explicitly parameterises berm geometry, it is, of the methods considered here, the closest in form to a free-standing rock berm. However, it was developed and validated for a toe structure that sits below and supports a breakwater armour slope, with wave run-down from that slope acting on the toe; it therefore assumes a slope configuration above the berm that is not present for a cable berm on a bed. Its applicability to the seabed types in this thesis should be treated as indicative rather than direct (Section 5.8).

$$Dn50 = 0.32 \cdot H_s / \Delta (Nod^{\frac{1}{3}}) \cdot \left(\frac{B_t}{H_s}\right)^{0.1} \cdot \left(\frac{t_t}{H_s}\right)^{\frac{1}{3}} \cdot \left(\frac{\hat{u}}{\sqrt{(g \cdot H_s)}}\right)^{\frac{1}{3}}$$

Variable	Meaning	How to determine
Nod	Damage number (displaced stones / D_{n50} width)	Chosen design value
h_t	Water depth above the toe/berm (m)	Bathymetry
B_t	Berm width (m)	Geometry input
t_t	Berm thickness (m)	Geometry input
$\hat{u}$	Characteristic orbital velocity at the toe (m/s)	Linear wave theory, from H_s , $T_{m-1,0}$, h
$T_{m-1,0}$	Spectral wave period (s)	From wave spectrum / NKT data

5.5 Herrera & Medina (2015)

Herrera and Medina (2015) developed a design formula specifically for toe berms in very shallow water on steep sea bottoms (slope $m = 1/10$), under breaking wave conditions, precisely the situation for which Herrera and Medina show that existing formulae, including Van Gent and Van der Werf (2014), do not perform reliably. Their formula (Eq. 12) predicts the damage number Nod for a standard toe berm geometry ($B_t = 3 \cdot D_{n50}$, $t_t = 2 \cdot D_{n50}$), accompanied by 90% confidence intervals. Four damage levels are defined: $Nod < 0.5$ (no significant movement), $Nod = 1.0$ (significant movement, recommended as a reasonable design criterion), $Nod = 2.0$ (moderate damage, berm still supporting the armour), and $Nod = 4.0$ (failure).

The method is valid within the range:

$$\begin{aligned} -0.15 < h_s/H_{s0} < 1.5 \\ -0.5 < h_s/D_{n50} < 5.01 \\ 0.02 < s_{0p} \text{ (wave steepness)} < 0.07 \end{aligned}$$

for the standard berm geometry and slope $m = 1/10$ only. Because it is one of the few methods in this comparison explicitly validated for breaking waves on a steep bottom, it is directly relevant to the bedrock case in this thesis, particularly where the cable route lies within or close to the wave-breaking zone discussed in Chapter 2.

$$Nod \approx \left(\frac{(H_{s0} \cdot L_{0p})^{0.5}}{\Delta D_{n50}} - 5.5 \right) \cdot \left[\left(-0.2 \cdot \frac{h_s}{D_{n50}} + 1.4 \right) \cdot \exp \left(0.25 \cdot \frac{h_s}{D_{n50}} - 0.65 \right) \right]^{\frac{1}{0.15}}$$

Variable	Meaning	How to determine
H_{s0}	Deep-water significant wave height (m)	Offshore wave data
L_{0p}	Deep-water wavelength at $T_p = g \cdot T_p^2 / 2\pi$ (m)	From T_p
h_s	Water depth at the toe (m)	Bathymetry
Δ	Relative buoyant density, $(\rho_s - \rho_w) / \rho_w$	$\rho_s = 2650$ or 3000 kg/m^3 (variable)

5.6 Deltares HaSPro Handbook (2023)

The Deltares HaSPro Handbook (2023) represents the current state of the art for offshore scour and cable protection design, and is, of the five core sources, the only one developed explicitly for cable-cover berms. It distinguishes three separate performance requirements that a rock protection must satisfy: external stability (resistance to hydraulic loading), interface stability (preventing the underlying sand from being winnowed through the protection layer), and flexibility (the ability of the protection to accommodate bed-level changes without losing effectiveness).

For external stability, the handbook introduces a rock berm deformation model (RoBeD) that predicts the expected deformation of a berm directly, rather than only a required stone size. Reshaping becomes relevant for higher mobility numbers, above a Shields number of roughly 0.06, and the model defines four deformation classes (0 to 3), from no deformation (class 0, mobility below a threshold value everywhere) to complete wash-away of the profile (class 3, mobility above the threshold even at the flat seabed). Note: the handbook also publishes fixed mobility-limit tables, but these are specific to monopile scour protection (parameterised by pile diameter and KC-number) and are not directly transferable to a cable berm; for a berm, the deformation model itself is intended to be calculated case-by-case rather than read from a single fixed threshold value.

Because it is calibrated for offshore conditions and assumes a wave-current direction perpendicular to the structure, and because it explicitly addresses berm geometry, interface stability, and flexibility together, it is the most directly applicable design reference for the framework developed in this thesis (Chapter 7), alongside Van Rijn (2019) for the underlying stone-size calculation.

Beyond the qualitative deformation classes, the handbook provides an explicit calculation procedure for the required Top of Product Cover, TOPC, the installed rock cover needed so that, after reshaping under the design hydraulic exposure, the remaining cover does not fall below the Minimum Top of Cover, MTOC. The ratio between the bottom width and height of the reshaped (parabolic) berm profile is given empirically, fitted from physical model tests, as a function of the Shields mobility parameter θ :

$$B_{\text{berm}}/h_{\text{berm}} = f(\theta) = 6000\theta^2 + 50\theta + 2.4$$

The installed and minimum berm heights are related to the cover requirements via a cable-position offset, here denoted Δ_{cable} to avoid confusion with the relative density Δ used elsewhere in this thesis:

$$h_{\text{berm}} = \text{TOPC} + \Delta_{\text{cable}},$$

$$h_{\text{min}} = \text{MTOC} + \Delta_{\text{cable}}, \text{ with } \Delta_{\text{cable}} = D_{\text{cable}} \text{ for a surface-laid cable,}$$

$$\text{or } \Delta_{\text{cable}} = Z_{\text{DOL}} - Z_{\text{SBL}} \text{ for a free-spanning/buried cable.}$$

The cross-sectional area of the installed (trapezoidal) berm and of the minimum reshaped (parabolic) berm are:

$$A_{berm} = h_{berm} \cdot W_{TOP} + h_{berm}^2 \cdot \alpha$$

$$A_{min} = (2/3) \cdot B_{berm} \cdot h_{min}$$

Setting the installed area equal to the minimum reshaped area gives a quadratic equation for h_{berm} , which, after taking the maximum root and substituting the relations above, yields the required installed berm height and, finally, the required TOPC:

$$h_{berm} = 0.5 \cdot \left[\sqrt{((W_{TOP}/\alpha)^2 + (4/\alpha)(2/3) \cdot f(\theta) \cdot h_{min}^2) - \frac{W_{TOP}}{\alpha}} \right]$$

$$TOPC = 0.5 \cdot \left[\sqrt{(W_{TOP}/\alpha)^2 + (4/\alpha)(2/3) \cdot f(\theta) \cdot (M_{TOC} + \Delta_{cable})^2 - W_{TOP}/\alpha} \right] - \Delta_{cable}$$

$$f(\theta) = 6000\theta^2 + 50\theta + 2.4$$

Variable	Meaning	How to determine
θ (θ_c)	Shields mobility parameter for the design hydraulic exposure	$\theta_c = \tau_c / [(\rho_s - \rho_w) \cdot g \cdot d_{50}]$
TOPC	Required Top of Product Cover, i.e. installed rock cover (m)	Output
M _{TOC}	Minimum Top of Cover, minimum acceptable cover after reshaping (m)	Project requirement / input
W _{TOP}	Top width of the installed berm (m)	Geometry input
α	Side slope parameter, slope expressed as 1: α	Geometry input
Δ_{cable}	Cable-position offset (m)	D _{cable} for surface-laid; ZDOL - ZSBL for buried/free-spanning
D _{cable}	Outer diameter of the cable (m)	Cable specification

5.7 Rock Density

Rock density, $\Delta = (\rho_s - \rho_w)/\rho_w$, is treated in this thesis as an explicit design variable rather than a fixed material property, comparing a normal-density preset ($\rho_s = 2650 \text{ kg/m}^3$) against a high-density preset ($\rho_s = 3000 \text{ kg/m}^3$). The Deltares HaSPro Handbook (2023) shows that higher-density rock is less mobile than normal-density rock of the same grading, because the same mass is achieved with smaller stones that present a smaller area to the hydrodynamic loading; using its RoBeD deformation model, Deltares finds that a normal-density ($\rho_s \approx 2650 \text{ kg/m}^3$) OSG45/180mm grading and a smaller OSG22/125mm grading achieve a similar degree of deformation once the density of the smaller grading exceeds roughly 3000 kg/m^3 .

This has a direct, practical consequence for the optimisation objective of this thesis: a higher rock density can allow a smaller stone size, and potentially a smaller layer thickness and total rock volume, for the same accepted level of damage. Since the design procedure itself (Sections 5.1–5.6) does not change with density, only Δ , comparing designs on normal- and high-density presets is treated in this thesis as a direct trade-off between material availability (normal-density rock is generally easier to source) and total tonnage, rather than as two separate design methods.

5.8 Comparative Summary

Method	Situation it was made for	Required inputs	Output	Damage criterion	Validity limits	Fit for nearshore cable berm
Hudson (1958)	Slope armour, rubble-mound, non/mild breaking	Hs, KD, Δ , cota	W50 / Dn50	Implicit fixed (~0–5%), no scale	Slope armour; no Tp, no current; small dataset	Low – first estimate only
Van der Meer (1988/98)	Slope armour, random waves	Hs, Tm (sm), Δ , Dn50, cota, permeability P, N waves, Sd	Dn50 / Ns at given Sd	$S_d = Ae/Dn50^2$ (start ≈ 2 , failure $\approx 8-17$)	Slope armour, not depth-limited; no current	Low–med – slope, not berm/bed
Van Rijn (2019)	Horizontal/mild-slope bed protection, current $\pm$ waves	uc, Hs, Tp, h, Δ , slope, $\alpha(ks)$, r, γ_{str}	required D50; + Nmr/Sd (Cheng)	$r = 0.4-1.0 \rightarrow \theta_{cr} = r-0.05$; Sd / moving rocks	D50 = 0.03–3 m; horizontal/mild slope; sparse field calibration	High – only one with combined wave+current; static & dynamic
Van Gent & Van der Werf (2014)	Toe of breakwater armour slope	Hs, Tm-1,0, ht, Bt, tt, Dn50, Δ , cota, roughness	Nod or Dn50	Nod (displaced stones / Dn50 width)	ht/h = 0.10–0.30; h/Hs = 1.2–4.5; slopes 1:1.5 & 1:2; waves only	Med – berm geometry fits, but assumes armour slope above & down-rush
Herrera & Medina (2015)	Toe berm, very shallow + steep (m=1/10), breaking	hs, Hs0, L0p (Tp), Δ , Dn50 (std Bt=3Dn50, tt=2Dn50)	Nod (Eq. 12) + 90% CI	Nod (0.5 start, 1 design, 2 moderate, 4 failure)	$-0.15 < hs/Hs0 < 1.5$; $-0.5 < hs/Dn50 < 5.01$; $0.02 < s0p < 0.07$; m=1/10; std berm only	Med–high for bedrock/steep breaking; fixed geometry, waves only
Deltares HaSPro / RoBeD (2023)	Loose-rock scour protection & cable-cover berms, offshore, current+waves	Shields θ ($\rightarrow$ uc, Hs, Tp, h, Δ , D50), WTOP, MTOC, Dcable, side slope	berm geometry (Bberm/hberm, TOPC) + deformation class; external/interf ace/flexibility	MOBthresh, case-specific, no fixed default; classes 0–3; reshaping $\theta > -0.06$	reshaping regime; perpendicular wave+current assumed; offshore-calibrated	High – only one explicitly for cable berms + 3 requirements

Situation	Primary method	Supporting / check	Why
Bedrock, steep nearshore, breaking waves	Herrera & Medina (2015)	Van Gent (range check); Hudson (first estimate)	Only method for steep bottom + breaking; demonstrates that gentle-slope formulas fail here.
Mobile sand, combined waves + current (crossing/obstacle)	Van Rijn (2019)	Deltares RoBeD (deformation + interface/flexibility)	Only method with genuine wave+current on a bed; Deltares covers winnowing/falling apron on sand.
Cable berm geometry & reshaping/deformation (any)	Deltares RoBeD (2023)	Van Rijn (D50 sizing)	Provides berm geometry + deformation class rather than only D50.
Quick first sizing / sanity check	Hudson / Van der Meer	—	Classical and fast, but without wave period or current.

- Schiereck & Verhagen (2019)

Ch. 8.2 (Stability general)

Ch. 8.3 (Stability of loose grains, slope stability under wave loading)

- Van Rijn (2019) – full paper
- Van Gent & Van der Werf (2014/ICCE2014) – full paper
- Herrera & Medina (2015) – full paper
- Deltares Handbook (2023)

Ch. 4 (Design of loose rock scour protections)

Ch. 5 (Loose rock berms at cable crossings)

6. Scour

6.1 The Scour Process

Scour is the local or large-scale erosion of the seabed caused by the flow disturbance introduced by a structure, or by natural morphodynamic processes. Two spatial scales are distinguished. Local scour develops around individual structural elements, and around discrete, blunt obstructions such as piles it is characteristically driven by a horseshoe vortex that forms where the approaching flow separates at the base of the structure, together with flow acceleration and wake turbulence, it typically develops relatively quickly. A low, sloped structure such as a rock berm does not present the same blunt, vertical face, so this specific mechanism is less relevant there. Edge scour around a berm (Section 6.3) is instead driven by flow acceleration and locally amplified turbulence at its boundary. Global scour, or global bed level lowering, is a broader depression around an entire structure or footprint, or results from large-scale morphodynamics unrelated to any single structure. It generally takes considerably longer to develop than local scour, and the two are not mutually exclusive. A structure can show both a local scour hole and a wider global lowering around it at the same time (Deltares HaSPro Handbook, 2023, Ch. 2).

Scour development also depends on whether sediment is supplied from upstream. In clear-water scour, there is no sediment transport into the scour hole from upstream, either because the upstream velocity is below the threshold of motion or because the upstream bed is not erodible. The scour hole develops relatively slowly and its final depth is sensitive to the exact flow conditions. In live-bed scour, sediment is supplied from upstream but the local transport capacity around the structure is larger, so material is removed faster than it is replenished, live-bed scour reaches its equilibrium depth considerably faster than clear-water scour, though the depth of the hole then fluctuates as bedforms migrate through it (Schiereck & Verhagen, 2019, Ch. 4). The maximum scour depth generally occurs near the transition between the two regimes, and this transition is what the empirical scour-depth formulas in the literature are calibrated to predict.

6.2 Scour Around Subsea Cables

Most scour research and design guidance, including the depth predictions referred to above, has been developed for compact structures such as monopiles, jacket legs, or breakwater heads. A subsea cable is physically a very different object: elongated and narrow relative to its length, lying directly on or just below the seabed rather than penetrating it as a pile does. Flow disturbance around a cable is therefore closer to that around a long cylinder lying on the bed than to that around a vertical pile. A horseshoe vortex can still form where flow separates in front of the cable, similar to a pipeline, but it develops along the length of the cable rather than as a single localised feature. The resulting scour pattern is a trench-like depression that can run continuously along sections of the route rather than as a localised pit at a single point.

The direct consequence for structural integrity is that scour beneath a cable can cause a section of cable to become unsupported, or free-spanning, over the resulting depression. A free span introduces bending and, where currents or waves generate oscillating flow past the span, the possibility of vortex-induced vibration and fatigue loading that a fully supported cable does not experience. Because a rock berm protecting a cable is itself an

elongated structure with the same alignment as the cable, edge scour along the berm (Section 6.3) and the free-spanning behaviour of the cable underneath it are closely related, rather than two separate problems.

6.3 Scour and Berm stability

Once a section of seabed is protected by a rock berm, scour does not stop, it shifts to the edges of the protected area. These edge scour holes form because the presence of the berm itself locally amplifies the flow and increases turbulence at its boundary, in a manner similar to local and global scour around an unprotected structure. Edge scour holes are generally smaller than local scour holes around an unprotected structure and take longer to develop, but in a tidal environment they tend to develop on alternating sides of the berm as the current reverses, since each reversal partially backfills the hole formed during the previous phase (Deltares HaSPro Handbook, 2023, Ch. 2).

This has a direct consequence for berm design: if the edges of a rock berm are undermined by scour, the berm can lose its foundation support at the margins and settle or slump into the resulting hole, regardless of whether the armour stones on top of the berm are individually stable under the design wave and current loading. Scour and berm stability are therefore coupled and cannot be designed independently: a berm sized correctly for hydraulic stability (Chapter 5) can still fail if its edges are not adequately protected against, or given sufficient allowance for, ongoing edge scour. Extending the berm in the direction of the dominant flow, so that a larger area of erosion-resistant material covers the zone where edge scour would otherwise develop, is one way this coupling is addressed in practice.

6.4 Scour Mitigation Approaches

Loose rock protection, in the form of a graded rock berm, is the primary scour and cable protection method considered in this thesis, and is treated in detail in Chapter 5. The Deltares HaSPro Handbook (2023) also describes several alternative scour protection systems that are used in practice as either an alternative to, or in combination with, loose rock:

- Artificial vegetation systems consist of a dense canopy of individual synthetic fronds attached to an anchored frame of ballasted mattresses. They work by damping near-bed turbulence and reducing flow velocities at the bed rather than by armouring it directly, and their design considerations include both overall stability and the prevention of edge scour around the mattress frame.
- Block, gabion, and ballast-filled mattresses are weighted mattress systems placed directly on the seabed. Block mattresses use interlocking concrete units, gabion mattresses use rock- or gravel-filled wire mesh compartments, and ballast-filled mattresses use a flexible fabric envelope filled with a ballast material; all three are used to protect pipelines and cables as well as structures, and, like loose rock, remain susceptible to edge scour at their perimeter if that is not separately addressed.

Each of these alternative systems has different installation requirements, flexibility to accommodate seabed movement, and edge-scour behaviour, and the Deltares Handbook indicates suitability per situation rather than a single preferred system. For the scope of this thesis, loose rock berms remain the primary method considered, since they are the most established and most directly comparable across the five core sources; the alternative systems above are noted here as context for situations where a loose rock berm may not be the preferred or feasible solution.

6.5 Anchor and Trawling Protection

Alongside hydrodynamic loading, a rock berm intended to protect a subsea cable may also need to withstand impact loads from bottom-trawling fishing gear and dragging ship anchors. These loads are not computed as part of the design tool in this thesis, since including them by default would risk significant overdesign. They are treated here only as boundary conditions to be aware of. It should also be stressed that the required level of protection against these man-made hazards is not a fixed, universal value: it is determined per project through a site-specific risk assessment (a Cable Burial Risk Assessment, following DNV and CarbonTrust guidance), based on the actual shipping density, fishing intensity, and anchorage activity along that particular section of route. The figures given below are therefore illustrative of common industry practice, not fixed design rules that apply everywhere.

For trawling gear, industry practice (Van Kessel & Stam, 2010; CIRIA, 2007) has shown that a minimum rock cover of 0.50 m over the cable is sufficient to protect against dragging trawl boards and beams in effectively all cases, with typical trawl board weights of 500–2000 kg at speeds of 3–5 knots corresponding to impact energies of 0.5–6 kNm; the slope of the berm deflects the trawl board so that only part of this energy needs to be absorbed, and penetration into the berm remains below about 0.30 m. This 0.5 m figure is a widely used project baseline in areas with active bottom trawling, but the actual required cover is still assessed per project against local fishing gear and activity.

Protection against a dragging ship anchor is a different matter, and even more strongly project- and location-dependent. Van Kessel and Stam (2010) show that rock berm dimensions designed to force a large anchor to break out and re-surface, rather than cut into or under the berm, scale directly with anchor size: for a representative 3-tonne stockless anchor, their rules of thumb already indicate a berm with a bottom width of around 15 m. Because anchor-related berm requirements can become disproportionately large compared to the wave-driven design considered in this thesis, computing anchor protection by default would directly conflict with the "avoid overdesign" objective of this thesis. Consistent with industry guidance, the recommended approach where large dragging anchors pose a credible risk is to route the cable away from high-risk areas (shipping lanes, anchorages, fishing grounds) rather than to design the berm for the largest conceivable anchor impact; anchor protection is therefore noted here as a boundary condition, to be sized on a per-project basis according to the identified risk, and as a possible follow-up study, not as part of the computational design tool developed in Chapter 7.

- Van Kessel, R., Stam, C.J. (2010), "The Lights Go Out: The Ultimate Protection Technology for Protecting Submarine Cables", SubOptic 2010.
- CIRIA (2007), The Rock Manual: The Use of Rock in Hydraulic Engineering, 2nd edition, CIRIA C683
- Schiereck & Verhagen (2019) Ch. 4 (Flow-Erosion: scour as sediment transport, scour process, scour with bed protection, equilibrium scour, stability of protection)
- Deltares Handbook (2023), Ch. 2 (Scour prediction), Ch. 3 (Scour mitigation strategies A/B/C), Ch. 5 (Loose rock berms at cable crossings), Ch. 9–11 (Alternative systems)
- Schiereck & Verhagen (2019), Ch. 4 (Flow-Erosion: scour as sediment transport, scour process, scour with bed protection, equilibrium scour, stability of protection.
- Deltares Handbook (2023), Ch. 2 (Scour prediction, general description of the scour process, global and local scour, edge scour), Ch. 9 (Alternative scour protection systems), Ch. 10 (Artificial vegetation), Ch. 11 (Block, gabion and ballast-filled mattresses)

7. Design Framework & Computational Approach

7.1 Input Parameter Space

All parameters required for a stability assessment, grouped by category: hydrodynamic (H_s , T_p , $T_m-1,0$, u_c , wave direction), geometric (h , h_t , bed slope), material (D_{n50} , ρ_s , D_{50}) and berm geometry (B_t , t_t). For each parameter, physical meaning, unit, and typical nearshore range.

7.2 Core Design Formulae

All formulae that will be implemented in the Python script, written out in full with variable definitions.

This chapter is the technical foundation of the script, every formula here maps directly onto the code.

Organization per method so its well-structured for the code. Like Chapter 6?

7.3 Failure Criteria & Damage Levels

Stability for each method.

Shields with a threshold ($\theta \leq \theta_{cr}$).

Van Rijn with damage classes ($r = 0.4$ to 1.0).

Van Gent NOD with an acceptable upper bound.

I have to make these differences explicit so they can be implemented consistently in the script.

7.4 Validity Ranges & Applicability Warnings

State the validity limits per method (e.g. Van Gent: $h_t/h = 0.10-0.30$, $h/H_s = 1.2-4.5$). In the script these become the warnings issued to the user when input falls outside the valid range. This is essential for a tool intended for global application.

7.5 Sensitivity Analysis Framework

Sensitivity analysis as a methodology: Identifying which input parameters most strongly influence the outcome.

Which methods do I want to apply

local sensitivity analysis and global methods such as Sobol indices and the Morris screening method.

Thesis methodology chapter.

- Schiereck & Verhagen (2019)

Ch. 3.2 (Stone dimensions, waterdepth, practical application)

Ch. 10 (Dimensions, probabilistics, failure mechanisms)

- Van Rijn (2019)

Section 2–4 (all core formulae and validity ranges)

- Van Gent & Van der Werf (ICCE2014)

Equations 1–3 and validity discussion

- Herrera & Medina (2015)

Equations 1–4 and validity ranges

- Deltares Handbook (2023)

Ch. 4.4 (Design approach, external stability, interface stability, flexibility)

Ch. 4.5 (Design workflow, design parameters)

Output must be for every little part of the cable route what options are there for constructing the cross-section. So if we use this d50 we need to use this slope, if we use this d50 we need to use this slope.

8. Knowledge Gaps & Research Questions

Conclusion of the literature review.

What do I know:

1. stability criteria for uniform flow are well developed,
2. the Deltares HaSPro approach is state-of-the-art for offshore applications,
3. and several methods exist for toe and berm stability.

What we do not know:

1. there is no validated, integrated method specifically for nearshore cable berms under combined wave-current conditions,
2. and existing methods are frequently applied outside their validity ranges without adequate justification.

I have to formulate the research questions that follow directly from the literature, which justify my methodology chapter and the development of my Python tool.

Rather than treating the six stability methods as six competing options to choose between, I've organized the tool around three stages that reflect what each type of method is actually used for.

Stage one is obtaining a candidate rock size. Hudson, Van der Meer, Van Rijn, and Van Gent & Van der Werf all belong here, each takes the site conditions and gives back a required stone size, at some assumed level of accepted damage.

Stage two is comparing that rock size against the level of damage it implies, and making an actual decision. Rather than just accepting whatever single answer the tool produces, the idea is to show the trade-off, how the required stone size changes as the accepted damage level changes, across a range, so the choice of "how conservative do we want to be here" is visible and deliberate, not hidden inside a default value. Herrera & Medina fits here specifically, since its formula doesn't give a stone size directly at all, it gives you damage for a candidate size, so you only get an answer by exploring exactly this kind of trade-off.

Stage three is finalizing the design with the Deltares HaSPro/RoBeD model. Once a rock size and an accepted damage level have been chosen, Deltares takes that and produces the actual buildable berm, height, required cover, and whether interface stability and flexibility hold up over time.

I think this three-stage structure is a useful way to present the design framework in the thesis, it makes clear that the tool isn't just running six formulas and picking one, it reflects a genuine engineering decision process, size it, evaluate the trade-off, finalize it.

Where this structure runs into a real problem:

The issue sits exactly between stage two and stage three. Each method expresses "acceptable damage" differently, an implicit fixed level for Hudson, a damage number S_d for Van der Meer, a reduction coefficient r for Van Rijn, a damage number N_{od} for Van Gent and Herrera & Medina, and deformation classes 0 through 3 for Deltares. These aren't the same physical quantity, and there's no established formula or table converting one into another, they come from different test setups and different structures.

So whatever damage level someone accepts in stage two is expressed in that method's own language, but stage three, Deltares, needs a deformation class, not an r or a N_{od} . There's currently no justified way to carry the decision from stage two into stage three, without either inventing an unbacked conversion, or asking for a second, disconnected decision about acceptable deformation class that has no explicit link to what was already decided.

This is already flagged as an open question in my project scope, mapping r , N_{od} , and deformation class onto a common axis is something still to be agreed, not something I've solved. My options as I see them:

- Option 1: I could just keep the two decisions separate and not try to link them. I pick my rock size using whichever method applies, and I'm happy with the damage level that method already uses. Then, separately, when I get to Deltares, I just choose a deformation class myself, without calculating it from the first choice. Two independent decisions, not one flowing into the other.
- Option 2: I could come up with my own rough conversion between the two, but say clearly that it's my own proposal, not something from the literature. So I'd say something like, "I'm treating this damage level as roughly equivalent to this deformation class," based on my own comparison, and in the thesis I'd be upfront that this needs validation later, it's a contribution I'm making, not a fact I found in a paper.
- Option 3: I could only connect the methods where a link genuinely makes sense, and leave the rest disconnected. So I'd connect Van Rijn to Deltares, since we already know those two are meant to work together for the sand case, but for a method like Herrera & Medina, where the link would be forced, I'd just present its result on its own, without trying to feed it into Deltares at all.

https://www.researchgate.net/publication/235634121_Experimental_Investigation_of_Water_Wave_Characteristics_in_a_Wave_Channel/figures?lo=1